

4. The Manipulator Jacobian.

Relationship between "Joint angles" and "End-effector state"

4.1 End-effector velocity.

Recall "Joint space" Q . "Forward kinematics map" $g_{st} : Q \rightarrow SE(3)$.
 base $\uparrow$
 to-L $\rightarrow$

eg: $Q = \underbrace{S' \times S'}_{\substack{\downarrow \\ \text{revolute} \\ \theta_1}} \times \underbrace{IR \times S'}_{\substack{\rightarrow \\ \text{prismatic} \\ A_3}}$

$$g = \begin{pmatrix} R & P \\ 0 & 1 \end{pmatrix}_{4 \times 4}$$

chain rule

$$\hat{V}_{st}^s = \frac{d}{dt} (g_{st}(\theta(t))) g_{st}^{-1}(\theta(t)) = \sum_{i=1}^n \left(\frac{\partial}{\partial \theta_i} g_{st} \dot{\theta}_i \right) g_{st}^{-1} = \sum_{i=1}^n \left(\frac{\partial g_{st}}{\partial \theta_i} g_{st}^{-1} \right) \dot{\theta}_i$$

$$\Rightarrow V_{st}^s = J_{st}^s(\theta) \cdot \dot{\theta}, \text{ with } J_{st}^s = \begin{bmatrix} \left(\frac{\partial g_{st}}{\partial \theta_1} g_{st}^{-1} \right)^V & \dots & \left(\frac{\partial g_{st}}{\partial \theta_n} g_{st}^{-1} \right)^V \end{bmatrix}$$

velocity of joint i .
 $\downarrow$
 6×1

Recall. $V = \begin{bmatrix} v \\ w \end{bmatrix}$ $\hat{V} = \begin{bmatrix} \hat{w} & v \\ 0 & 0 \end{bmatrix}$
 $\begin{bmatrix} \hat{w} & v \\ 0 & 0 \end{bmatrix}^V = \begin{bmatrix} v \\ w \end{bmatrix}_{6 \times 1}$

"Exponential formula" $g_{st}(\theta) = e^{\hat{\xi}_1 \theta_1} \dots e^{\hat{\xi}_n \theta_n} g_{st}(0)$ $\rightarrow$ initial configuration

$$\Rightarrow \frac{\partial g_{st}}{\partial \theta_i} g_{st}^{-1} = e^{\hat{\xi}_1 \theta_1} \dots e^{\hat{\xi}_{i-1} \theta_{i-1}} \frac{\partial}{\partial \theta_i} (e^{\hat{\xi}_i \theta_i}) \dots e^{\hat{\xi}_n \theta_n} g_{st}(0) g_{st}^{-1}$$

relate to the base frame

$$= e^{\hat{\xi}_1 \theta_1} \dots e^{\hat{\xi}_{i-1} \theta_{i-1}} \left(\frac{\partial}{\partial \theta_i} e^{\hat{\xi}_i \theta_i} \right) e^{\hat{\xi}_i \theta_i} \dots e^{\hat{\xi}_n \theta_n} g_{st}(0) g_{st}^{-1} e^{-\hat{\xi}_n \theta_n} \dots e^{-\hat{\xi}_1 \theta_1}$$

$$= e^{\hat{\xi}_1 \theta_1} \dots e^{\hat{\xi}_{i-1} \theta_{i-1}} \hat{\xi}_i e^{-\hat{\xi}_{i-1} \theta_{i-1}} \dots e^{-\hat{\xi}_1 \theta_1}$$

$$\Rightarrow \left(\frac{\partial g_{st}}{\partial \theta_i} g_{st}^{-1} \right)^V = \text{Ad} (e^{\hat{\xi}_1 \theta_1} \dots e^{\hat{\xi}_{i-1} \theta_{i-1}}) \xi_i$$

$$J_{st}^s(\theta) = [\xi_1, \xi_2, \dots, \xi_n]$$

with $\xi_i' = \text{Ad} (e^{\hat{\xi}_1 \theta_1} \dots e^{\hat{\xi}_{i-1} \theta_{i-1}}) \xi_i$

revolute joint

$$\xi_i = \begin{bmatrix} -w_i \times q_i \\ w_i \end{bmatrix}$$

prismatic joint

$$\xi_i = \begin{bmatrix} v_i \\ 0 \end{bmatrix} \rightarrow \text{direction of translation}$$

Recall $V_{st}^b = g_{st}^{-1} \frac{d}{dt} (g_{st}(\theta(t)))$

$$\Rightarrow J_{st}^b = \left[\left(g_{st}^{-1} \frac{\partial g_{st}}{\partial \theta_i} \right)^V \right] = \text{Ad}^T (e^{\hat{\xi}_1 \theta_1} \dots e^{\hat{\xi}_n \theta_n} g_{st}(0)) \xi_i$$

$$\Rightarrow J_{st}^s(\theta) = \text{Ad} (g_{st}(\theta)) J_{st}^b(\theta)$$

Recall $V_{st}^b = g_{st}^{-1}(\theta(t)) \frac{d}{dt} (g_{st}^*(\theta(t)))$
 $= \left[\left(g_{st}^{-1} \frac{\partial g_{st}}{\partial \theta_i} \right)^V \right] \dot{\theta}.$

$\Rightarrow$ Body manipulator Jacobian $J_{st}^b = \left(g_{st}^{-1} \frac{\partial g_{st}}{\partial \theta_i} \right)^V.$

By exponentials formula.

$$g_{st}^{-1} \frac{\partial g_{st}}{\partial \theta_i} = g_{st}^{-1}(0) \cdot e^{-\hat{\xi}_n \theta_n} \dots e^{-\hat{\xi}_i \theta_i} e^{\hat{\xi}_i \theta_i} \dots e^{\hat{\xi}_{i-1} \theta_{i-1}} \hat{\xi}_i e^{\hat{\xi}_i \theta_i} \dots e^{\hat{\xi}_n \theta_n} g_{st}(0).$$

$$= g_{st}^{-1}(0) \cdot e^{-\hat{\xi}_n \theta_n} \dots e^{-\hat{\xi}_i \theta_i} \hat{\xi}_i e^{\hat{\xi}_i \theta_i} \dots e^{\hat{\xi}_n \theta_n} g_{st}(0).$$

$$\left(g_{st}^{-1} \frac{\partial g_{st}}{\partial \theta_i} \right)^V = \text{Ad} \left(g_{st}^{-1}(0) e^{-\hat{\xi}_n \theta_n} \dots e^{-\hat{\xi}_i \theta_i} \right) \hat{\xi}_i$$

$$J_{st}^b = \text{Ad}^{-1} \left(e^{\hat{\xi}_i \theta_i} \dots e^{\hat{\xi}_n \theta_n} g_{st}(0) \right) \hat{\xi}_i.$$

$$\Rightarrow J_{st}^s(\theta) = \text{Ad} \left(g_{st}(\theta) \right) J_{st}^b(\theta).$$

The instantaneous velocity of a point.

①. q^b — point on the end-effector. (in body (tool) coordinates).

$$V_q^b = \hat{V}_{st}^b q^b = (J_{st}^b(\theta) \dot{\theta})^\wedge q^b.$$

②. q^s — point — — — end-effector (in spatial (base frame)).

$$V_q^s = \hat{V}_{st}^s q^s = (J_{st}^s(\theta) \dot{\theta})^\wedge q^s.$$

$$\bar{q}^s = g_{st}(\theta) \bar{q}^b \quad \bar{q}^b = \begin{bmatrix} q^b \\ 1 \end{bmatrix}, \quad \bar{q}^s = \begin{bmatrix} q^s \\ 1 \end{bmatrix}$$

If $q^b = 0$. $q^s = P(\theta) = \cancel{g_{st}(\theta)} q$ — the position component

of the forward kinematics map.

$$\begin{bmatrix} q^s \\ 1 \end{bmatrix} = \begin{bmatrix} R(\theta) & P(\theta) \\ 0 & 1 \end{bmatrix} \begin{bmatrix} q^b \\ 1 \end{bmatrix}$$

$$\Rightarrow \hat{V}_q = \begin{bmatrix} \dot{P}(\theta) \\ 0 \end{bmatrix} = \cancel{R_{st} \hat{V}_{st}^b} \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$R_{st} = \begin{bmatrix} R & 0 \\ 0 & 1 \end{bmatrix}$$

$$\hat{V}_{st}^b = \begin{bmatrix} \hat{w}^b & v^b \\ 0 & 0 \end{bmatrix}$$

$$\bar{q}^s = R(\theta) \bar{q}^b + P(\theta)$$

$$= P(\theta)$$

$$= \hat{V}_{st}^s \begin{bmatrix} P(\theta) \\ 1 \end{bmatrix} = \cancel{R_{st} \hat{V}_{st}^b} \begin{bmatrix} P(\theta) \\ 1 \end{bmatrix}$$

$$\bar{q}^b = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\hat{V}_{st}^b \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} \hat{w}^b [0] + v^b \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} v^b \\ 0 \end{bmatrix} \text{ BICMR}$$

$$= \cancel{R_{st} \hat{V}_{st}^b} R_{st}^{-1} g_{st} \hat{V}_{st}^b g_{st}^{-1} \begin{bmatrix} P(\theta) \\ 1 \end{bmatrix}$$

"The instantaneous velocity of a point"

① q^b — point on the end-effector (tool coordinates)

$$V_q^b = \hat{V}_{st}^b q^b = (J_{st}^b(\theta) \dot{\theta})^{\wedge} q^b.$$

② q^s — point on the end-effector (base ~~for~~ coordinates)

$$V_q^s = \hat{V}_{st}^s q^s = (J_{st}^s(A) \dot{\theta})^{\wedge} q^s.$$

If $q^b = 0$, (origin of the tool coordinate)

$p(\theta) = q^s$ (the ^{position} component)

$$g_{\tau}(\theta) = \begin{pmatrix} R(\theta) & P(\theta) \\ 0 & 1 \end{pmatrix} \Rightarrow \begin{bmatrix} q^s \\ 1 \end{bmatrix} = \begin{bmatrix} R(\theta) & P(\theta) \\ 0 & 1 \end{bmatrix} \begin{bmatrix} q^b \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} R(\theta)q^b + P(\theta) \\ 1 \end{bmatrix}$$

$$\Rightarrow V_q^s = \frac{d}{dt} \begin{bmatrix} q^s \\ 1 \end{bmatrix} = \begin{bmatrix} \dot{\theta} \\ 0 \end{bmatrix}$$

$$q_{st}^{-1} \begin{bmatrix} R & P \\ 0 & 1 \end{bmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \Rightarrow q^s = R(\theta)q^b + P(\theta) = P(\theta)$$

$$= \hat{V}_{st}^s q^s = \hat{V}_{st}^s \begin{bmatrix} p(\omega) \\ 1 \end{bmatrix}$$

$$\Rightarrow g_{st}^{-1} \begin{bmatrix} p \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$= g_{st} \hat{V}_{st}^b g_{st}^{-1} \begin{bmatrix} P(\theta) \\ 1 \end{bmatrix} = \underbrace{g_{st}}_{R_{st}} \hat{V}_{st}^b \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

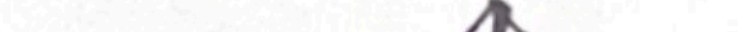
Equivalent expressions for the velocity of the origin of the tool frame.

⇒ "Relationship between joint velocity and end-effector velocity":

$$V_{st}^s(t) = J_{st}^s(\theta) \dot{\theta}(t)$$

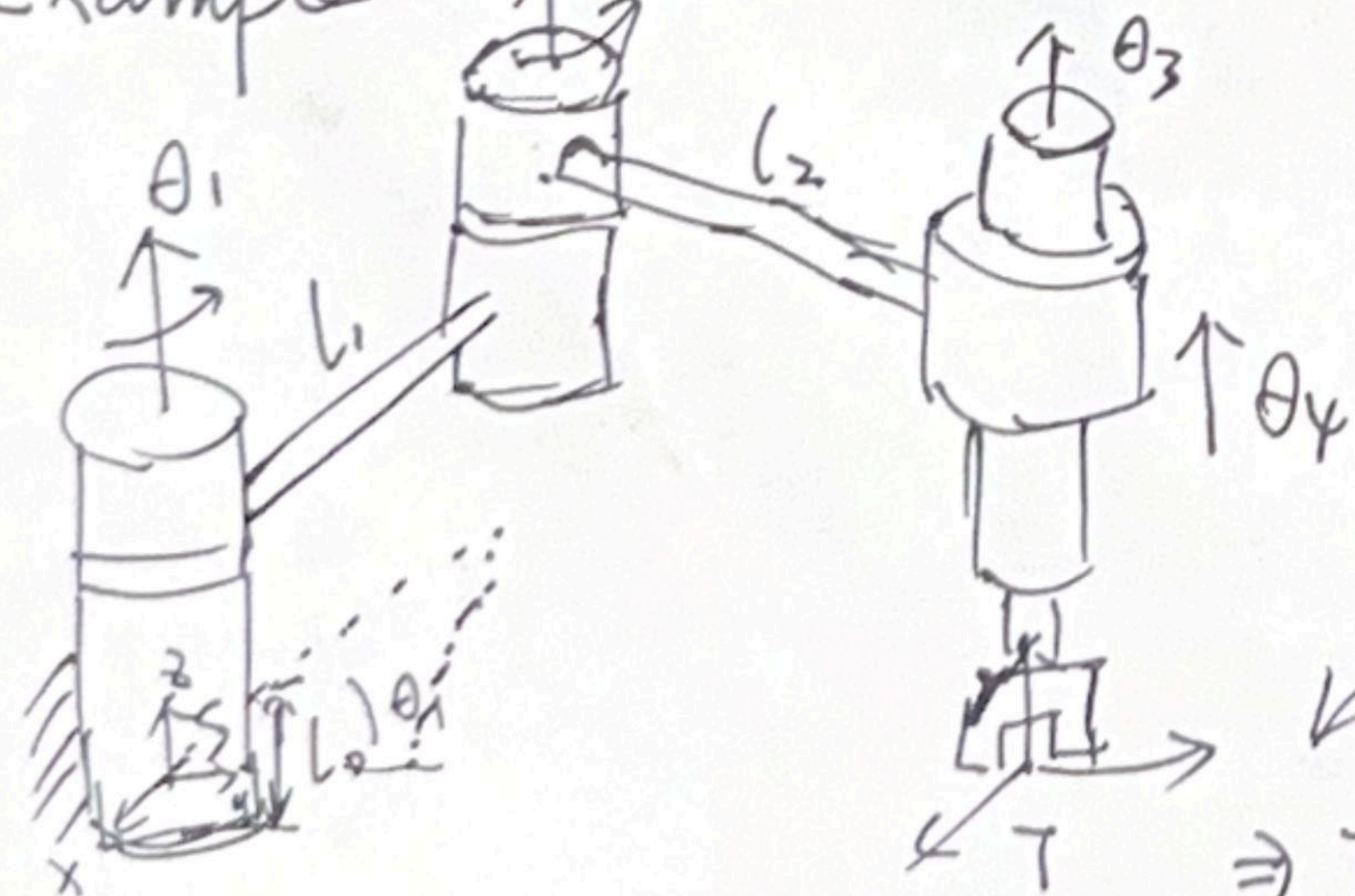
if J_{st} is invertible. $\Rightarrow \dot{\theta}(t) = (J_{st}^S(\theta))^{-1} V_{st}^S(t)$

"singular configuration"



$y = \begin{pmatrix} -g \times w \\ w \end{pmatrix}$

Example:  "Scara robot" w



$$w = (0, 0, 1) \quad q_1' = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \Rightarrow z_1 = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ -1 \end{pmatrix}$$

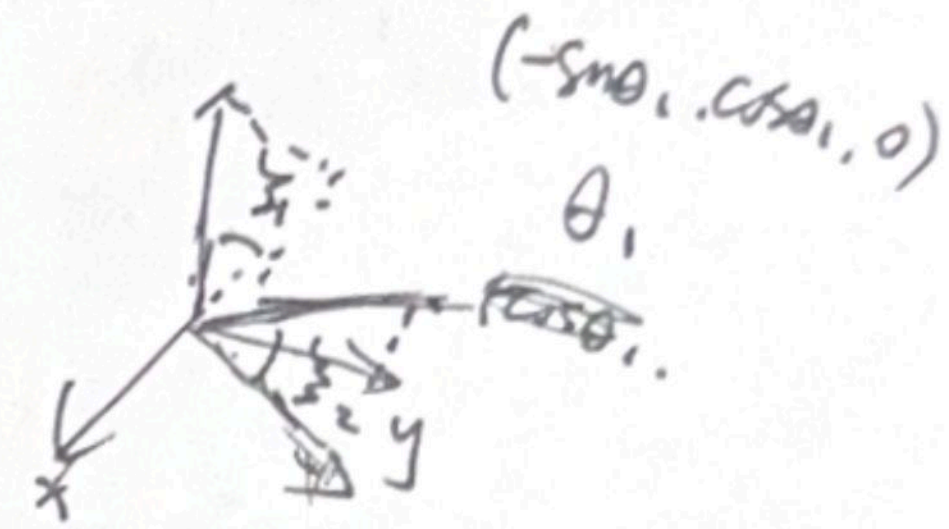
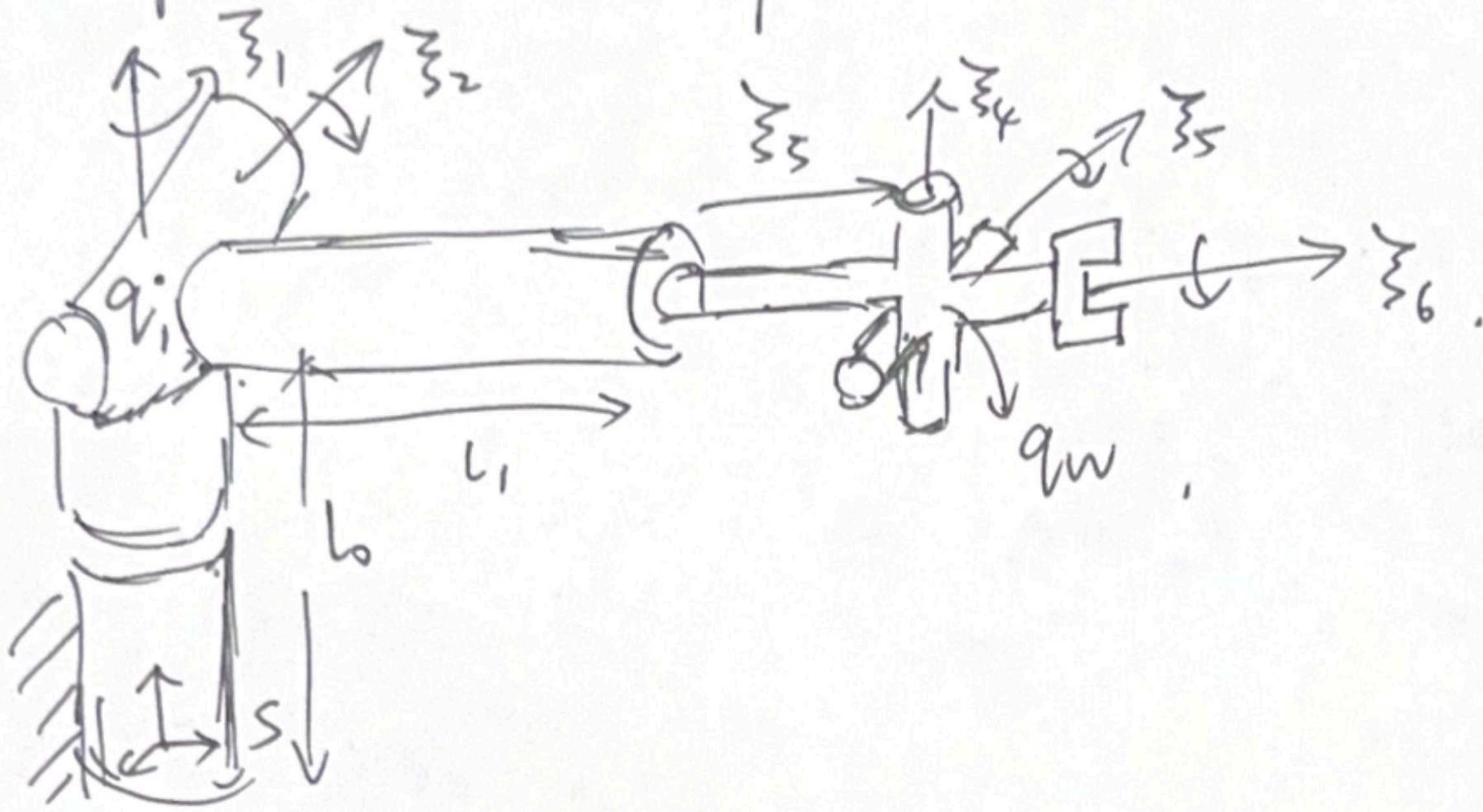
$$q_2 = \begin{pmatrix} -1 \sin \theta_1 \\ 1 \cos \theta_1 \\ 0 \end{pmatrix} \Rightarrow \sum_2 = \begin{pmatrix} 1 \cos \theta_1 \\ 1 \sin \theta_1 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

$$q_3 = \begin{pmatrix} -L_1 \sin \theta_1 - L_2 \sin(\theta_1 + \theta_2) \\ L_1 \cos \theta_1 + L_2 \cos(\theta_1 + \theta_2) \\ 0 \end{pmatrix} \Rightarrow \dot{q}_3 = \begin{pmatrix} \dot{q}_3^1 \\ \dot{q}_3^2 \\ \dot{q}_3^3 \end{pmatrix}$$

$$\Rightarrow J_{St}^S = [\xi_1, \xi_2', \xi_3']$$

$$p(\theta) = \begin{pmatrix} -l_1 \sin \theta_1 & -l_2 \sin(\theta_1 + \theta_2) \\ l_1 \cos \theta_1 & l_2 \cos(\theta_1 + \theta_2) \\ 0 & 0 \end{pmatrix} \Rightarrow \dot{p}(\theta) = \begin{pmatrix} -l_1 \cos \theta_1 \cdot \dot{\theta}_1 & -l_2 \cos(\theta_1 + \theta_2) (\dot{\theta}_1 + \dot{\theta}_2) \\ -l_1 \sin \theta_1 \cdot \dot{\theta}_1 & -l_2 \sin(\theta_1 + \theta_2) (\dot{\theta}_1 + \dot{\theta}_2) \\ 0 & 0 \end{pmatrix}$$

Example: "Stanford arm".



$$q_1 = q_2 = (0, 0, l_0) \quad W_1 = (0, 0, 1) \quad W_2' = (-\cos \theta_1, -\sin \theta_1, 0)$$

$$\Rightarrow z_1 = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} \quad z_2' = \begin{bmatrix} l_0 \sin \theta_1 \\ -l_0 \cos \theta_1 \\ 0 \\ -\sin \theta_1 \end{bmatrix}$$

$$V_3 = (0, 1, 0) \quad z_3' = \begin{pmatrix} e^{\hat{z}_1 \theta_1} e^{-\hat{x} \theta_2} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \\ 0 \end{pmatrix} = \begin{bmatrix} -\sin \theta_1 \cos \theta_2 \\ \cos \theta_1 \cos \theta_2 \\ -\sin \theta_2 \\ 0 \\ 0 \\ 0 \end{bmatrix} \approx \begin{bmatrix} V_3' \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$q_w' = l_1 \cdot V_3' + \dot{\theta}_3 V_3 + q_1 = \begin{pmatrix} -(l_1 + \theta_3) \sin \theta_1 \cos \theta_2 \\ (l_1 + \theta_3) \cos \theta_1 \cos \theta_2 \\ l_0 - (l_1 + \theta_3) \sin \theta_2 \end{pmatrix}$$

$$W_4' = e^{\hat{z}_1 \theta_1} e^{-\hat{x} \theta_2} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} \cdot \\ \cdot \\ \cdot \end{bmatrix} \quad W_5' = e^{\hat{z}_1 \theta_1} e^{-\hat{x} \theta_2} \begin{bmatrix} -1 \\ 0 \\ 0 \end{bmatrix}$$

$$W_6' = e^{\hat{z}_1 \theta_1} e^{-\hat{x} \theta_2} \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \Rightarrow z_4' = \begin{pmatrix} -W_4' \times q_w' \\ W_4' \end{pmatrix} \quad z_5' = \begin{pmatrix} -W_5' \times q_w' \\ W_5' \end{pmatrix}$$

$$z_6' = \begin{pmatrix} -W_6' \times q_w' \\ W_6' \end{pmatrix}$$

$$\Rightarrow J_{st}^s = [z_1, z_2', z_3', z_4', z_5', z_6']$$

4.2. End-effector Forces.

$$W = \int_{t_1}^{t_2} V_{st}^b \cdot F_t dt = \int_{t_1}^{t_2} \dot{\theta} \cdot \tau dt.$$

→ body wrench.

$$\Rightarrow \dot{\theta} \cdot \tau = V_{st}^b \cdot F_t = J_{st}^b \dot{\theta} \cdot F_t$$

$$\Rightarrow \dot{\theta}^T \tau = \dot{\theta}^T (J_{st}^b)^T F_t \Rightarrow \tau = (J_{st}^b)^T F_t.$$

Joint torques.

Similar. $\tau = (J_{st}^s)^T F_s.$

Example "Scara robot".

$$(J_{st}^s)^T = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 1 \\ l_1 c_1 & l_1 s_1 & 0 & 0 & 0 & 1 \\ l_1 c_1 + l_2 c_{12} & l_1 s_1 + l_2 s_{12} & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 & 0 \end{pmatrix}.$$

$$\Rightarrow F_{N_1} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, F_{N_2} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \text{ s.t. } \tau = 0.$$

4.4 Manipulability.

①. ability to reach a certain position. — range of pos.

②. ability to change the position. — ~~flexibility~~ velocity.

$$\mu_1(\theta) = \sigma_{\min}(J(\theta)). \quad \mu_2(\theta) = \frac{\sigma_{\min}(J(\theta))}{\sigma_{\max}(J(\theta))}.$$

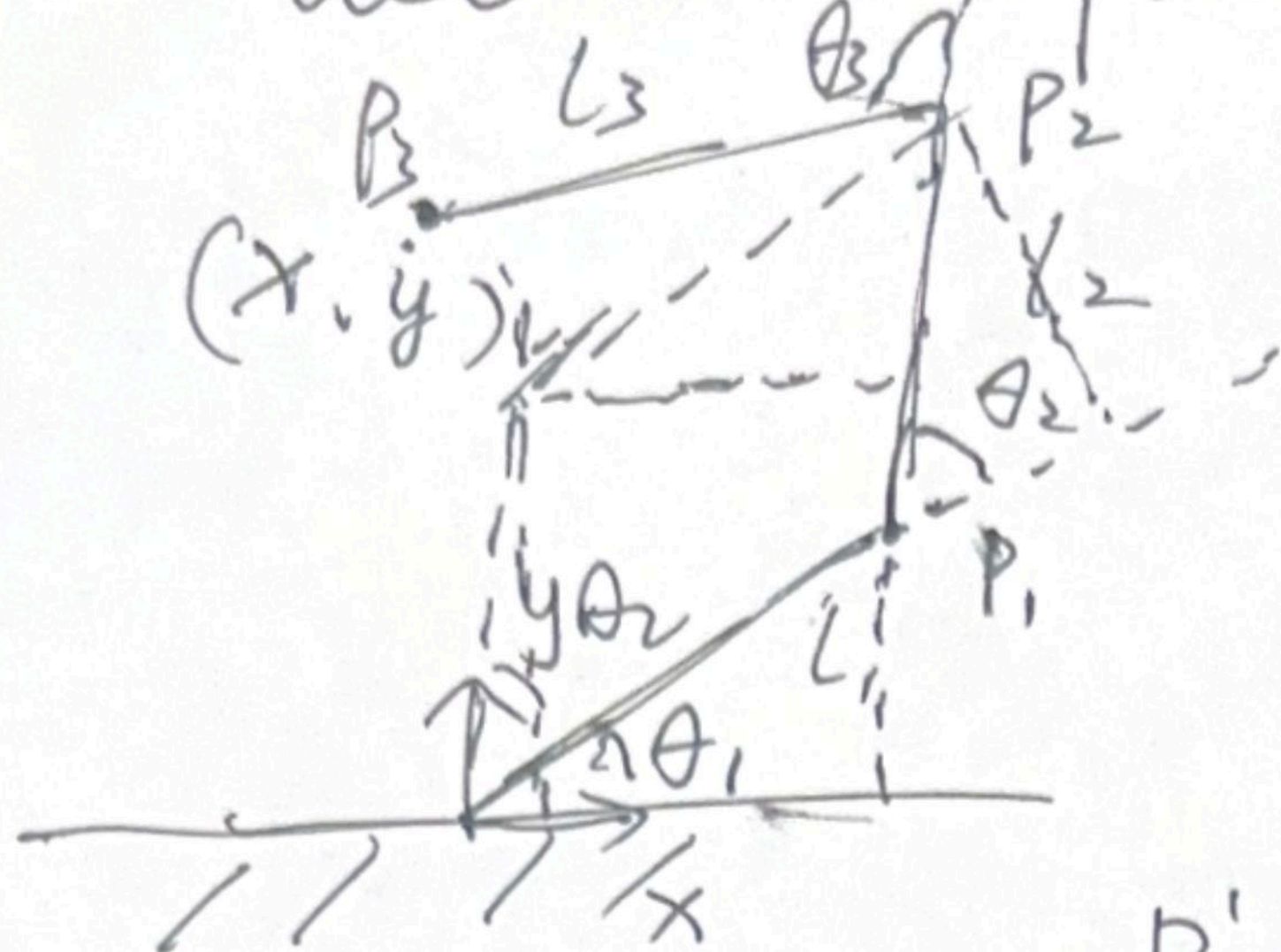
→ velocity.

$$\mu_3(\theta) = \det J(\theta).$$

↓
~~Sensitivity~~ average.

volume of the velocity.

Parallel Manipulators



Example 1.

$$P_1 = (l_1 \cos \theta_1, l_1 \sin \theta_1)$$

$$P_2' = (l_1 \cos \theta_1 + l_2 \cos (\theta_1 + \theta_2), l_1 \sin \theta_1 + l_2 \sin (\theta_1 + \theta_2))$$

$$P_3' = (l_1 \cos \theta_1 + l_2 \cos (\theta_1 + \theta_2) + l_3 \cos (\theta_1 + \theta_2 + \theta_3), l_1 \sin \theta_1 + l_2 \sin (\theta_1 + \theta_2) + l_3 \sin (\theta_1 + \theta_2 + \theta_3))$$

$$\frac{\partial P}{\partial \theta} = \begin{bmatrix} D \\ 0 \end{bmatrix}$$



"planar manipulator".

Example 2

$$g_{st} = e^{\hat{\xi}_1 \theta_1} \dots e^{\hat{\xi}_n \theta_n} g_E(0) = e^{\hat{\xi}_{2n} \theta_{2n}} \dots e^{\hat{\xi}_1 \theta_1} g_E(0)$$

degrees of freedom.

$$F = 6N - \sum_{i=1}^g (6 - f_i) \quad \mathbb{R}^3$$

$$F = 3N - \sum_{i=1}^g (3 - f_i) \quad \mathbb{R}^2$$

Example. "Four bar linkage."